

3.6.3 Skeletal muscles are stimulated to contract by nerves and act as effectors (A-level only)

Content	Opportunities for skills development
Muscles act in antagonistic pairs against an incompressible skeleton. Gross and microscopic structure of skeletal muscle. The ultrastructure of a myofibril. The roles of actin, myosin, calcium ions and ATP in myofibril contraction. The roles of calcium ions and tropomyosin in the cycle of actinomyosin bridge formation. (The role of troponin is not required.) The roles of ATP and phosphocreatine in muscle contraction. The structure, location and general properties of slow and fast skeletal muscle fibres.	AT d Students could examine prepared slides of skeletal muscle using an optical microscope. AT h Students could investigate the effect of repeated muscular contraction on the rate of muscle fatigue in human volunteers.